Signals and Systems
SET 214
Lecture (08)- Fourier Series

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Outlines

- Introduction to Fourier Series
- Exercise
- Trigonometric Fourier series
- Exercise
- Fourier Coefficients of Symmetrical Waveforms
- Exponential Fourier series
- Exercise
- Trigonometric Vs Exponential Fourier series

Outlines

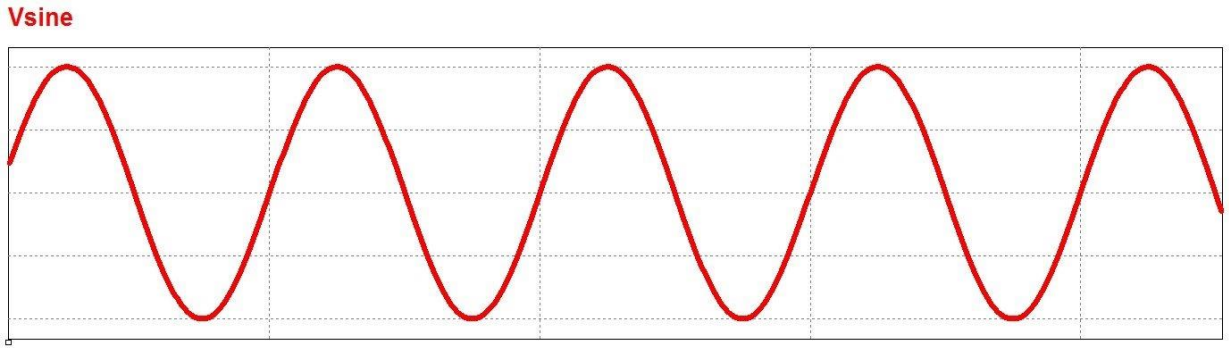
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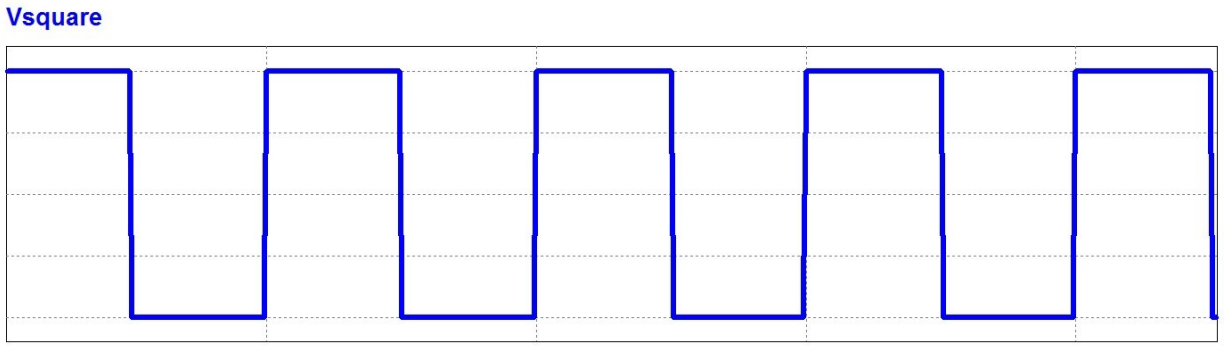
Introduction to Fourier Series (1/3)

- Most of the phenomena studied in the domain of Engineering and Science are periodic in nature.
- For instance, current and voltage in an alternating current circuit.
- These periodic functions could be analyzed into their constituent components (fundamentals and harmonics) by a process called Fourier Analysis.
- Periodic functions occur frequently in the problems studied during Engineering education. Their representation in terms of simple periodic functions such as sine and cosine function, which leads to Fourier series(FS)
- A Fourier Series is an approximate technique which can convert any non sinusoidal signal so that all periodic signals can be represented in unique form.
- It is useful for analyzing periodic signals only.

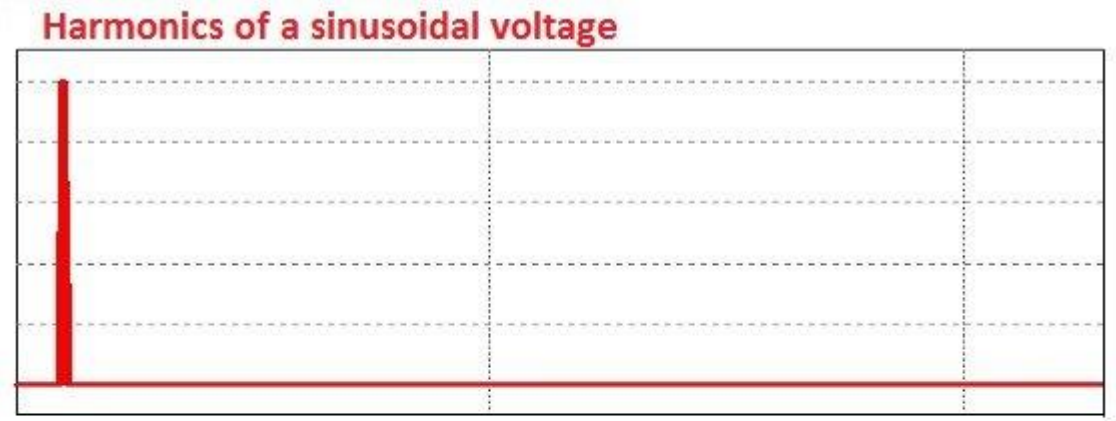
Introduction to Fourier Series (2/3)

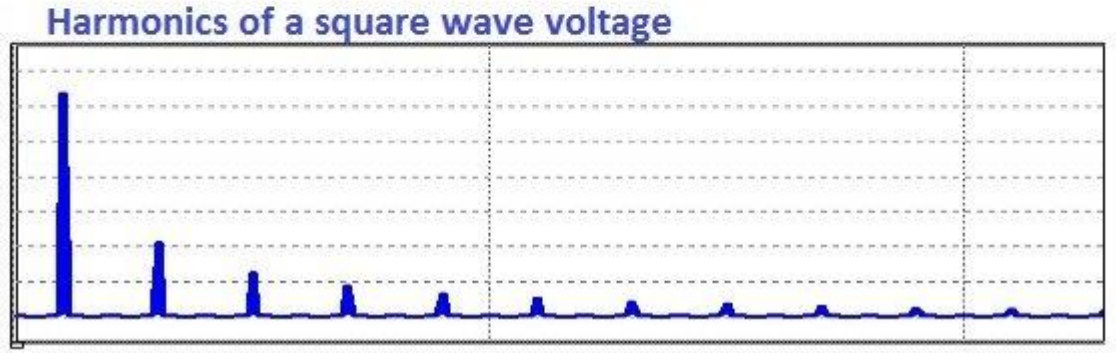
- It is easy to see the harmonic when examining the time domain and frequency domain representations of a square wave





A sinusoidal voltage and a square wave voltage in the time domain

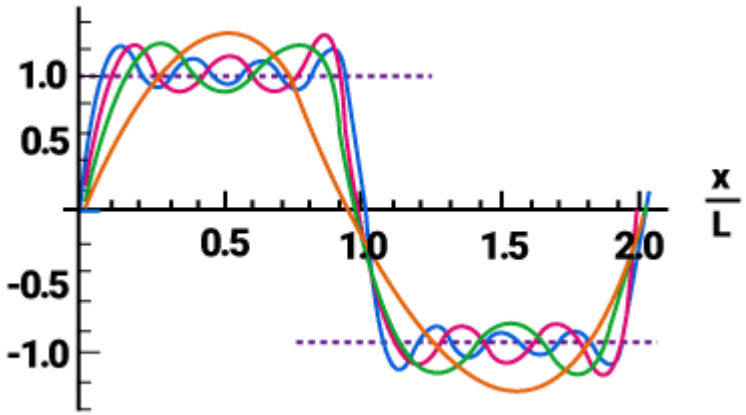




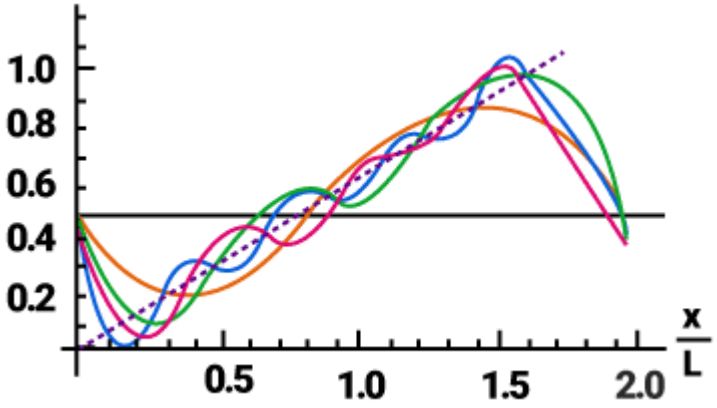
A sinusoidal voltage and a square wave voltage in the frequency domain; only the square wave has peaks at the harmonic frequencies.

Introduction to Fourier Series (3/3)

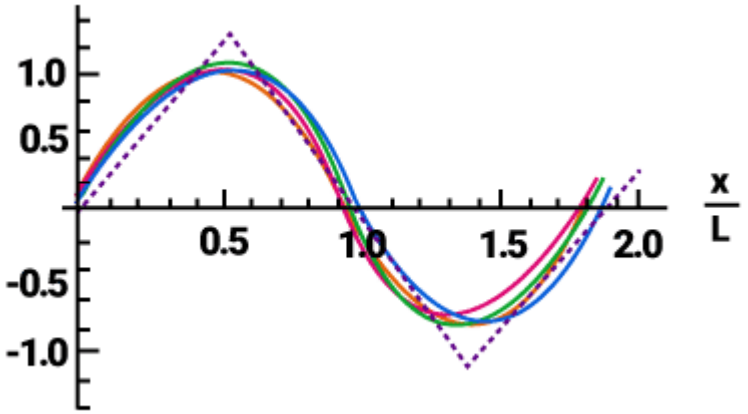
- What is the Fourier Series?
  - A Fourier series is an expansion of a periodic function f(x) in terms of an infinite sum of sines and cosines. Fourier Series makes use of the orthogonality relationships of the sine and cosine functions.



SQUARE WAVE



SAWTOOTH WAVE



TRIANGLE WAVE

- Types of Fourier series
  - Trigonometric Fourier series
  - Exponential Fourier series

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Exercise

❑ Determine whether the functions given can be represented by a Fourier series?

$$X(t)=\cos 6t+\sin 8t$$

Ans.

$$T1=\frac{2\pi}{6}=\frac{\pi}{3}$$

$$T2=\frac{2\pi}{8}=\frac{\pi}{4}$$

$$\frac{T1}{T2}=\frac{\pi/3}{\pi/4}=\frac{4}{3}$$

Rational number, so periodic so, it can be represented by Fourier series

NOTE

FS is for analyzing periodic signals only.

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Trigonometric Fourier series

- Any function can be expressed as a (infinite) linear combination of sines and cosines.
- Any function x(t) in the interval (t_o, t_o+T) can be represented as:-

$$x(t) = \underbrace{\frac{a_o}{2}}_{\text{DC Part}} + \underbrace{\sum_{n=1}^{\infty} a_n Cos(nw_o t)}_{\text{Even Part}} + \underbrace{\sum_{n=1}^{\infty} b_n \, sin(nw_o t)}_{\text{Odd Part}}$$

$$a_o = \frac{2}{T}\int_{to}^{to+T} x(t)dt$$

$$a_n = \frac{2}{T}\int_{to}^{to+T} x(t)\, cos(nw_o t)dt \qquad \text{n=1,2,3,....}$$

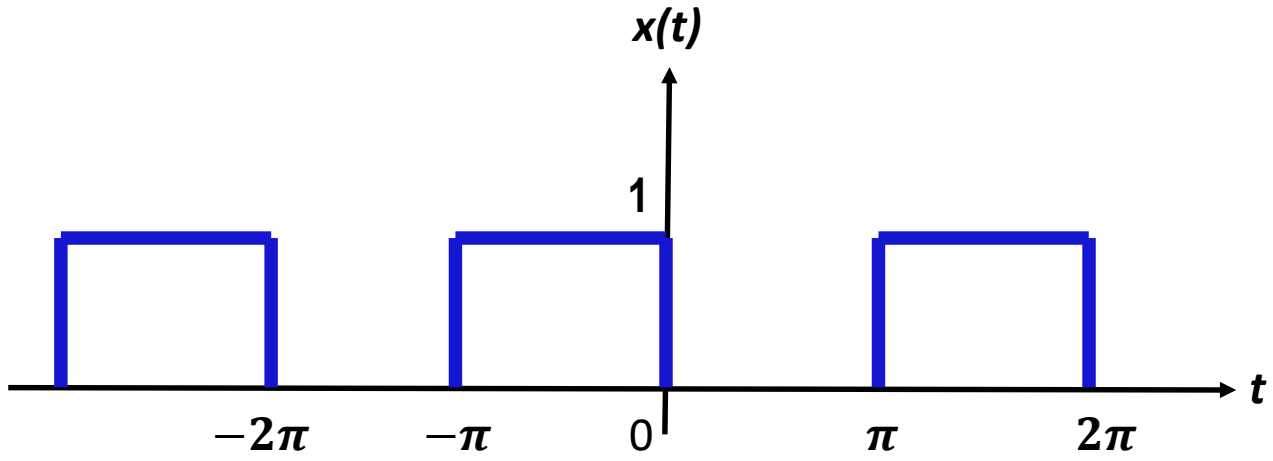
$$b_n = \frac{2}{T}\int_{to}^{to+T} x(t)\, sin(nw_o t)dt \qquad \text{n=1,2,3,....}$$

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Exercise

- Find the Fourier Series expansion of the periodic waveform given, where period (T) $=2\pi$



Exercise

Ans.

$$x(t) = \underbrace{\frac{a_0}{2}}_{\text{DC Part}} + \underbrace{\sum_{n=1}^{\infty} a_n \cos(n\omega_0 t)}_{\text{Even Part}} + \underbrace{\sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)}_{\text{Odd Part}}$$

❑Step(1):-

$$a_o = \frac{2}{T}\int_{to}^{to+T} x(t)dt$$

$$a_o = \frac{2}{2\pi}\int_{-\pi}^{\pi} x(t)dt = \frac{1}{\pi}\left[\int_{-\pi}^{0} 1\,dt + \int_{0}^{\pi} 0\,dt\right]$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} 1\,dt = \frac{1}{\pi}\,t\Big|_{-\pi}^{0} = \frac{1}{\pi}\,[0-(-\pi)]$$

$$= 1$$

❑Step(2):-

$$a_n = \frac{2}{T}\int_{to}^{to+T} x(t)\,cos(nw_o t)dt = \frac{2}{2\pi}\int_{-\pi}^{\pi} x(t)\,cos(nw_o t)dt$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} 1 * cos(nw_o t)dt + \int_{0}^{\pi} 0 * cos(nw_o t)dt\right]$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} 1 * cos(nw_o t)dt = \frac{1}{\pi}\left[\frac{sin(nw_o t)}{nw_o}\right]\Big|_{-\pi}^{0}$$

$$= \frac{1}{nw_o\pi}[sin(0) - sin(-nw_o\pi)]$$

$$= \frac{1}{nw_o\pi}[0 + sin(nw_o\pi)]$$

$$= \frac{1}{n\pi}[0 + sin(n\pi)]$$

$$= \frac{1}{n\pi}[0+0] = 0$$

$$w_o = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

Exercise

Ans.

❑Step(3):-

$$b_n = \frac{2}{T}\int_{to}^{to+T} x(t)\,sin(nw_o t)dt = \frac{2}{2\pi}\int_{-\pi}^{\pi} x(t)\,sin(nw_o t)dt$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} 1 * sin(nw_o t)dt + \int_{0}^{\pi} 0 * sin(nw_o t)dt\right]$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} 1 * sin(nw_o t)dt = \frac{1}{\pi}\left[\frac{-cos(nw_o t)}{nw_o}\right]\Big|_{-\pi}^{0}$$

$$= \frac{-1}{nw_o\pi}[cos(0) - cos(-nw_o\pi)]$$

$$= \frac{-1}{nw_o\pi}[1 - cos(-nw_o\pi)]$$

$$b_n = \frac{-1}{nw_o\pi}[1 - cos(nw_o\pi)] \quad \Rightarrow \quad w_o = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

$$= \frac{-1}{n\pi}[1 - cos(n\pi)]$$

$$b_n = \begin{cases} 0, & n \rightarrow even \\ \frac{-2}{n\pi}, & n \rightarrow odd \end{cases}$$

n	1	2	3	4
bn	-2/π	0	-2/3π	0

Substituting our results now gives the required series:-

$$x(t) = \underbrace{\frac{a_0}{2}}_{\text{DC Part}} + \underbrace{\sum_{n=1}^{\infty} a_n \cos(n\omega_0 t)}_{\text{Even Part}} + \underbrace{\sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)}_{\text{Odd Part}}$$

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\left(\frac{-2}{n\pi}\right)\sin(nt)$$

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Fourier Coefficients of Symmetrical Waveforms

- The use of symmetry properties simplifies the calculation of Fourier coefficients.

$$x(t) = \underbrace{\frac{a_0}{2}}_{\text{DC Part}} + \underbrace{\sum_{n=1}^{\infty} a_n \cos(n\omega_0 t)}_{\text{Even Part}} + \underbrace{\sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)}_{\text{Odd Part}}$$

- Even Functions $x(t) = x(-t)$
  - DC Part
  - Fourier series expansion of even signal does not contain sine terms

$$x(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t$$

$$a_n = \frac{4}{T}\int_0^{T/2} x(t)\cos(n\omega_0 t)dt$$

Odd Functions $x(t) = -x(-t)$

- No DC Part
- Fourier series expansion of odd signal contain sine terms only

$$x(t) = \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$$

$$b_n = \frac{4}{T}\int_0^{T/2} x(t)\sin(n\omega_0 t)dt$$

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Exponential Fourier series

$$x(t) = \frac{a_o}{2} + \sum_{n=1}^{\infty} a_n Cos(nw_o t) + \sum_{n=1}^{\infty} b_n \, sin(nw_o t)$$

$$a_o = \frac{2}{T}\int_{to}^{to+T} x(t)dt$$

$$a_n = \frac{2}{T}\int_{to}^{to+T} x(t)\, cos(nw_o t)dt \qquad \text{n=1,2,3,....}$$

$$b_n = \frac{2}{T}\int_{to}^{to+T} x(t)\, sin(nw_o t)dt \qquad \text{n=1,2,3,....}$$

$$x(t) = \sum_{n=-\infty}^{\infty} c_n \, e^{jnw_o t}$$

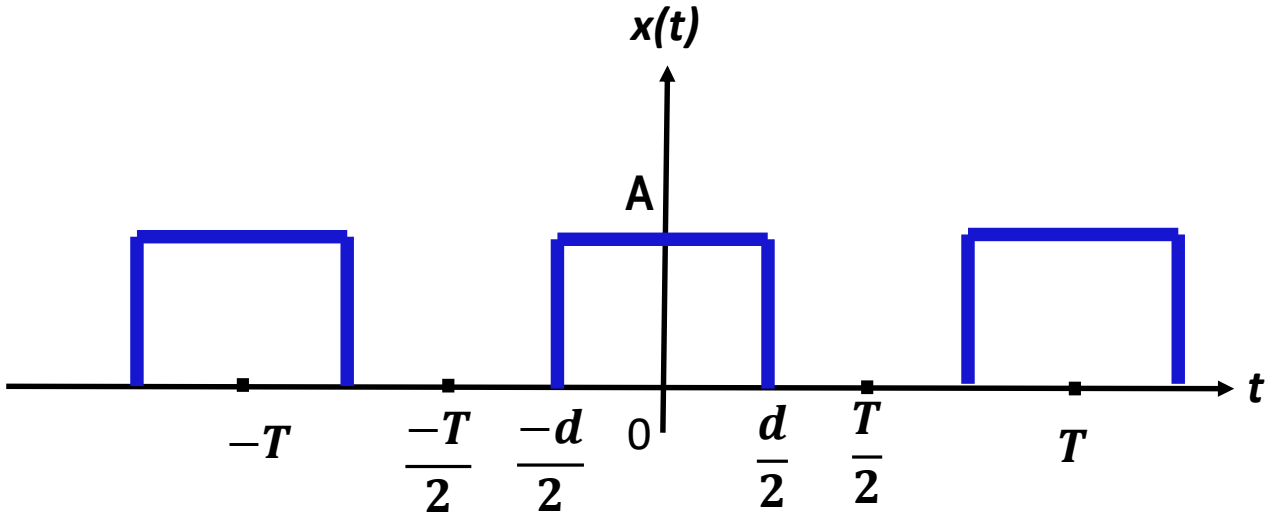
$$c_n = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{-jnw_o t}dt$$

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Exercise (1/2)

Q1) Find the Exponential Fourier Series of the periodic waveform given, where T=2π



Exercise (2/2)

Ans.

$$x(t) = \sum_{n=-\infty}^{\infty} c_n \, e^{jnw_o t}$$

$$c_n = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{-jnw_o t}dt$$

$$c_n = \frac{1}{T}\int_{-d/2}^{d/2} A\, e^{-jnw_o t}dt$$

$$c_n = \frac{A}{T}\,\frac{1}{-jnw_o}\, e^{-jnw_o t}\Big|_{-d/2}^{d/2}$$

$$c_n = \frac{A}{T}\,\frac{1}{-jnw_o}\left[\, e^{-jnw_o\frac{d}{2}} - e^{-jnw_o\frac{-d}{2}}\right]$$

$$c_n = \frac{A}{T}\,\frac{1}{-jnw_o}\left[\, e^{-jnw_o\frac{d}{2}} - e^{jnw_o\frac{d}{2}}\right]$$

$$c_n = \frac{A}{T}\,\frac{1}{jnw_o}\left[\, -e^{-jnw_o\frac{d}{2}} + e^{jnw_o\frac{d}{2}}\right]$$

$$c_n = \frac{2 * A}{T * nw_o}\,\frac{\left[\, e^{jnw_o\frac{d}{2}} - e^{-jnw_o\frac{d}{2}}\right]}{2j}$$

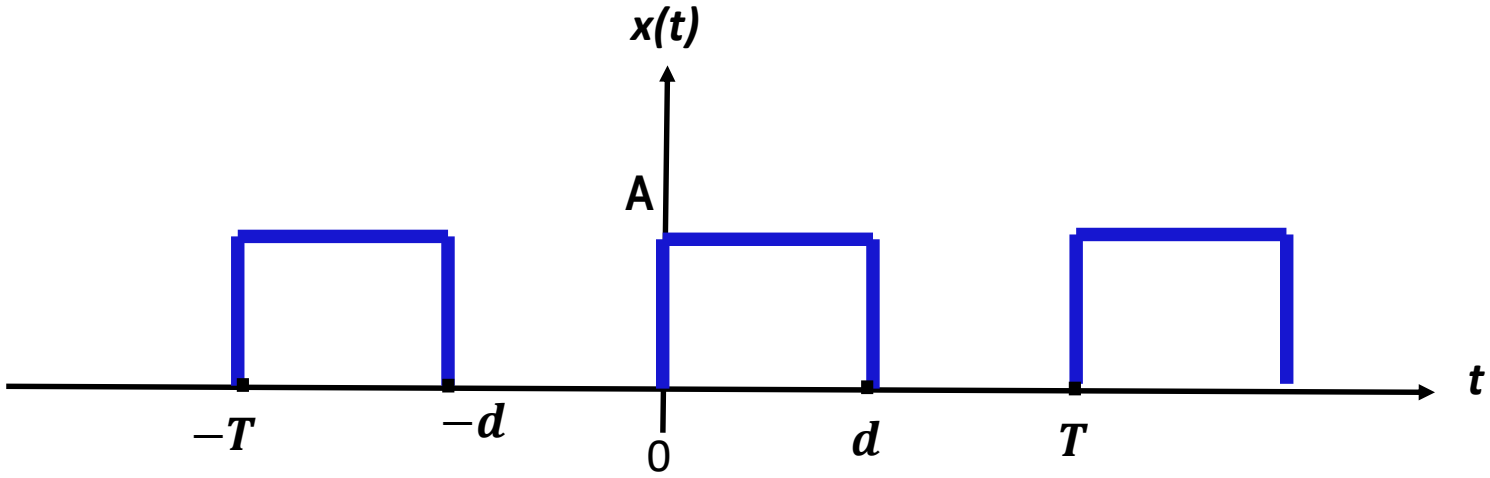
$$c_n = \frac{2 * A}{T * nw_o}\, sin(nw_o\frac{d}{2}) \quad \Rightarrow \quad where, w_o = \frac{2\pi}{T} = 1$$

$$c_n = \frac{A}{n\pi}\, sin(\,\frac{n\pi d}{T})$$

$$x(t) = \sum_{n=-\infty}^{\infty} \frac{A}{n\pi}\, sin\left(\,\frac{n\pi d}{T}\right) e^{jnt}$$

Exercise (1/2)

Q2) Find the Exponential Fourier Series of the periodic waveform given, where T=2π



Exercise (2/2)

Ans.

$$x(t) = \sum_{n=-\infty}^{\infty} c_n \, e^{jnw_o t}$$

$$c_n = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{-jnw_o t} dt$$

$$c_n = \frac{1}{T}\int_{0}^{d} A\, e^{-jnw_o t} dt$$

$$c_n = \frac{A}{T}\,\frac{1}{-jnw_o}\, e^{-jnw_o t}\Big|_0^d$$

$$c_n = \frac{A}{T}\,\frac{1}{-jnw_o}\,[\, e^{-jnw_o d} - e^{0}]$$

$$c_n = \frac{A}{T}\,\frac{1}{jnw_o}\,[\, -e^{-jnw_o d} + 1]$$

$$c_n = \frac{A}{T}\,\frac{1}{jnw_o}\, e^{-jnw_o \frac{d}{2}}\,[\, -e^{-jnw_o \frac{d}{2}} + e^{jnw_o \frac{d}{2}}]$$

$$c_n = \frac{A}{T}\,\frac{1}{jnw_o}\, e^{-jnw_o \frac{d}{2}}\,[\, e^{jnw_o \frac{d}{2}} - e^{-jnw_o \frac{d}{2}}]$$

$$c_n = \frac{2 * A}{T * nw_o}\, e^{-jnw_o \frac{d}{2}}\,\frac{[\, e^{jnw_o \frac{d}{2}} - e^{-jnw_o \frac{d}{2}}]}{2j}$$

$$c_n = \frac{2 * A}{T * nw_o}\, e^{-jnw_o \frac{d}{2}}\, sin(nw_o \frac{d}{2}) \quad where, w_o = \frac{2\pi}{T} = 1$$

$$c_n = \frac{A}{n\pi}\, e^{-jn\frac{d}{2}}\, sin(\frac{n\pi d}{T})$$

$$x(t) = \sum_{n=-\infty}^{\infty} \frac{A}{n\pi}\, e^{-jn\frac{d}{2}}\, sin\left(\frac{n\pi d}{T}\right) e^{jnt}$$

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Trigonometric Vs Exponential Fourier series (1/2)

Euler's Equations

$$e^{jnw_o t} = \cos(nw_o t) + j\sin(nw_o t)$$

$$e^{-jnw_o t} = \cos(nw_o t) - j\sin(nw_o t)$$

$$\cos(nw_o t) = \frac{1}{2}(e^{jnw_o t} + e^{-jnw_o t})$$

$$\sin(nw_o t) = \frac{1}{2j}(e^{jnw_o t} - e^{-jnw_o t})$$

$$x(t) = \frac{a_o}{2} + \sum_{n=1}^{\infty} a_n Cos(nw_o t) + \sum_{n=1}^{\infty} b_n \sin(nw_o t)$$

$$x(t) = \frac{a_o}{2} + \frac{1}{2}\sum_{n=1}^{\infty} a_n(e^{jnw_o t} + e^{-jnw_o t}) + \frac{1}{2j}\sum_{n=1}^{\infty} b_n(e^{jnw_o t} - e^{-jnw_o t})$$

$$x(t) = \frac{a_o}{2} + \frac{1}{2}\sum_{n=1}^{\infty} a_n(e^{jnw_o t} + e^{-jnw_o t}) - \frac{j}{2}\sum_{n=1}^{\infty} b_n(e^{jnw_o t} - e^{-jnw_o t})$$

$$x(t) = \frac{a_o}{2} + \sum_{n=1}^{\infty}[\frac{1}{2}(a_n - jb_n)e^{jnw_o t} + \frac{1}{2}(a_n + jb_n)e^{-jnw_o t}]$$

$$x(t) = C_o + \sum_{n=1}^{\infty}[c_n e^{jnw_o t} + c_{-n}e^{-jnw_o t}]$$

Trigonometric Vs Exponential Fourier series (2/2)

$$x(t) = \frac{a_o}{2} + \sum_{n=1}^{\infty} a_n Cos(nw_o t) + \sum_{n=1}^{\infty} b_n \, sin(nw_o t)$$

$$x(t) = \frac{a_o}{2} + \frac{1}{2}\sum_{n=1}^{\infty} a_n(e^{jnw_o t} + e^{-jnw_o t}) + \frac{1}{2j}\sum_{n=1}^{\infty} b_n(e^{jnw_o t} - e^{-jnw_o t})$$

$$x(t) = \frac{a_o}{2} + \frac{1}{2}\sum_{n=1}^{\infty} a_n(e^{jnw_o t} + e^{-jnw_o t}) - \frac{j}{2}\sum_{n=1}^{\infty} b_n(e^{jnw_o t} - e^{-jnw_o t})$$

$$x(t) = \frac{a_o}{2} + \sum_{n=1}^{\infty}[\frac{1}{2}(a_n - jb_n)e^{jnw_o t} + \frac{1}{2}(a_n + jb_n)e^{-jnw_o t}]$$

$$x(t) = C_o + \sum_{n=1}^{\infty}[c_n e^{jnw_o t} + c_{-n} e^{-jnw_o t}]$$

$$c_0 = \frac{a_0}{2}$$

$$c_n = \frac{1}{2}(a_n - jb_n)$$

$$c_{-n} = \frac{1}{2}(a_n + jb_n)$$

Thanks!